What should United have done in summer 2023?

Two alternative transfer windows from the model against the real one, decided on 2022-23 data and graded on 2023-24

Viktor Sadowski · October 2026 · case study to What is a Premier League player worth? · code: football-analytics

Summary

In summer 2023 Manchester United spent EUR 203m on fees for Mount, Onana, Højlund and Bayındır and took in EUR 61m from sales, a net spend of EUR 142m. This case asks what the model in the main report would have done with the same net spend, the same squad rules and only the data available that summer, and then grades the answer on what the players really did in 2023-24.

The model searches for two alternative windows, called scenario A and scenario B. Both start from United's real squad in June 2023, make at most six signings or sales, and can only buy players a club would realistically sell to United. Scenario A, best value, looks for the window that is worth the most money over the players' contracts, as long as United are projected at least as many points as if they had done nothing. Scenario B, most points, looks for the window that gives the most projected 2023-24 points, as long as it doesn't lose money over the contracts.

Scenario B (Bruno Guimarães, Theo Hernández, Udogie, Botman, Solly March and Sørloth in; Varane, Martial, Telles, Van de Beek, Malacia and Greenwood out, a net spend of EUR 129m) was projected to give United an 87% chance of the top four, against 40% for the real window. Scored with every player's real 2023-24 numbers, it comes to 77-78 league points, against the 60 United got. Scenario A sells for EUR 116m more than it buys and lands at 65-69 points. The real window comes out level with doing nothing. The model would not have started Mount or Højlund over the players United already had, and in 2023-24 Mount played 514 league minutes and Højlund rated close to replacement level (explained below).

How to read the numbers

The case uses a few terms from the main report. They are explained here so the rest can be read on its own.

- Value of a player: how much better his team does per 90 minutes when he is on the pitch, measured in non-penalty expected goals (npxG) for minus npxG against, compared with an average big-five player. It comes from RAPM, a regression on every stretch of play where the same 22 players were on the pitch. A value of +0.10 means his team creates 0.10 npxG more than it concedes per 90 minutes than with an average player. Over a full season (3,420 minutes) that is 3.8 npxG, which is worth about 2.8 league points.
- Replacement level: what a club gets from a player it can sign for next to nothing, measured on players in the bottom 20% of market values in their league. A whole team of replacement-level players would expect about 40 points in the Premier League, roughly 15th or 16th place and a few points above the relegation zone.
- Points above replacement: the league points a player adds over such a player in the minutes he is projected to play. For a squad it is the sum over the players. 60 points, United's real 2023-24 total, is about 7th place in an average season; 70 is usually 4th.
- Window value: the money a window adds over the contracts of the players involved, in EUR m. Each player's points are priced at EUR 4.1m a point (Premier League prize money per place), minus his wages, plus what he can be sold for later, and future years are discounted 8% a year. Fees paid and sales received are added, and standing still, doing nothing that summer, is the zero.

- Hindsight: the windows are chosen on data up to 2022-23. Afterwards every player gets the real value he had in 2023-24, and each window is scored again with those values.

1. The setup

Everything the model uses is cut at the end of 2022-23: player values, injury risk, minutes, club Elo ratings, and the fitted models for fees, resale and how long a player stays at a club. The rules for both scenarios:

- Budget: the net spend on fees can't be more than United's real net spend that summer, EUR 142m, so the scenarios compete with the real window on the same money. Wages are not in the budget, they are part of the window value.
- Prices: buying a player costs what the market would charge, from a fee model fitted on big-five summer transfers up to 2022. It uses his Transfermarkt market value on 1 June 2023, his age and the extra a Premier League buyer pays. Selling a United player brings in 80% of his market value, since a club that has to sell rarely gets full price.
- Squad rules: at most 25 senior players, of which at least 8 homegrown (three years at English or Welsh clubs between age 15 and 21, approximated from transfer history). Players born in 2002 or later are under 21 and don't count towards the 25.
- Track record: a player needs at least 1,500 minutes in big-five league matches over the three seasons up to 2022-23 to be a target. With fewer minutes his value is too uncertain to pay for.
- Sellers: a target must either play for a club with a lower Elo rating than United, or be surplus where he is (less than half of his club's minutes when fit, or out on loan). This removes 123 of the 1,405 targets with a track record, among them Pedri, Rodri, Upamecano, Nkunku and Araújo. The best 133 by position are kept as the shortlist.
- Minutes: United play a 4-2-3-1, split into ten positions (the two centre-backs share one). A player can fill his main position and any other he started at least 20% of his games in (and at least five games). The minutes of each position go to the best player first, capped at 92% of the season and his expected time out injured, and the next best covers the rest.
- Contracts and wages for United's own players: contract end dates as of summer 2023 from press lists, wages as reported in the press where there is a report (Casemiro, Bruno Fernandes, Rashford, Garnacho), else from the wage model in the main report.
- Search: at most six moves, each a signing, a sale or both. The search is greedy: each round it makes the single move that improves the scenario's goal the most, and it stops when no move helps.

United before the window

Standing still, the model projected United's squad at 37.6 points above replacement for 2023-24, which the simulator in section 3 turns into 69 expected league points. That was too high, as section 4 shows, but it is the line both scenarios have to beat. The weak spots by the model were central midfield (Eriksen at 31 with Fred and nobody behind them), the striker position (Martial, often injured) and left-back cover behind Shaw, who missed half of most seasons. It rated Sancho and Antony as United's best wide players and Rashford lower than his reputation, from the 2022-23 on/off numbers (section 6 of the main report has the caveats).

2. The two scenarios

The two scenarios answer different questions. Scenario A asks how much money the window can make without making United worse on the pitch, which is the classic moneyball question. Scenario B asks how much better United can get on the pitch without the window losing money over the contracts. Table 1 has the money and the projected points side by side.

	Fees paid (EUR m)	Sales (EUR m)	Net spend (EUR m)	Window value (EUR m)	Projected points above replacement
Standing still	0	0	0	0	37.6
Real window	203	61	+142	-244	35.7
Scenario A: best value	105	222	-116	+496	42.3
Scenario B: most points	253	125	+129	+108	49.0

Table 1. The money and projected 2023-24 points of each window, as seen in summer 2023. A negative net spend means more came in from sales than went out on fees. Window value is against standing still.

	Standing still	Real window	Scenario A: best value	Scenario B: most points
GK	Dean Henderson (Matej Kovar)	André Onana (Altay Bayındır)	Dean Henderson (Matej Kovar)	Dean Henderson (Matej Kovar)
LB	Luke Shaw (Tyrell Malacia)	Luke Shaw (Tyrell Malacia)	Iyenoma Destiny Udogie (Tyrell Malacia)	Theo Hernández (Iyenoma Destiny Udogie)
CB	Harry Maguire, Victor Lindelöf (Lisandro Martínez)	Harry Maguire, Victor Lindelöf (Lisandro Martínez)	Facundo Medina, Warmed Omari (Victor Lindelöf)	Victor Lindelöf, Sven Botman (Harry Maguire)
RB	Aaron Wan-Bissaka (Diogo Dalot)	Aaron Wan-Bissaka (Diogo Dalot)	Issa Kabore (Diogo Dalot)	Aaron Wan-Bissaka (Diogo Dalot)
DM	Casemiro (Fred)	Casemiro (Scott McTominay)	Casemiro (Fred)	Casemiro (Fred)
CM	Christian Eriksen (Fred)	Christian Eriksen (Kobbie Mainoo)	Christian Eriksen (Fred)	Bruno Guimarães (Christian Eriksen)
LW	Jadon Sancho (Alejandro Garnacho)	Jadon Sancho (Alejandro Garnacho)	Jadon Sancho (Alejandro Garnacho)	Jadon Sancho (Iyenoma Destiny Udogie)
AM	Bruno Fernandes (Christian Eriksen)	Bruno Fernandes (Christian Eriksen)	Bruno Fernandes (Christian Eriksen)	Bruno Fernandes (Christian Eriksen)
RW	Antony (Anthony Elanga)	Antony (Amad Diallo)	Bradley Barcola (Anthony Elanga)	Solly March (Antony)
FW	Anthony Martial (Marcus Rashford)	Anthony Martial (Marcus Rashford)	Alexander Sørloth (Mason Greenwood)	Alexander Sørloth (Marcus Rashford)

Table 2. Who plays where in each window, starters first and backups in brackets, from the minutes planned in summer 2023.

Scenario A trades starters on big contracts for cheaper players the model rates as good or better: Barcola for Antony, Medina for Martínez, Udogie for Rashford (Udogie plays left-back, Sancho keeps the left wing), Omari for Varane, Kaboré for Wan-Bissaka and Sørloth for Martial. It brings in EUR 116m more than it spends and saves a lot of wages, which is where most of its window value comes from. Scenario B fixes the spots the model saw as weak: Bruno Guimarães in central midfield, Theo Hernández at left-back with Udogie behind him, Botman next to Lindelöf, Solly March on the right and Sørloth up front. The real window spent most of its money on positions the model already had covered. Onana replaces Henderson in goal, a small step up. Mount and Højlund get no minutes in the model's plan at all: it rated Bruno Fernandes and Eriksen above Mount (+0.09), and Martial and Rashford above Højlund (+0.05), a 20-year-old striker it rated close to replacement level. Their EUR 148m in fees bought backups, which is why the real window projects slightly below standing still in Table 1.

3. The pre-season odds

A season simulator plays the real 2023-24 fixture list 20,000 times. Every club's strength is its squad on 1 September 2023, player values times projected minutes. United's strength is moved by the difference between each window and the real one. In each simulated season every club's strength also gets a random error, sized to match how wrong the August forecast has been in earlier seasons: about 10.5 points per club, of which 7.4 is the luck of single matches. Tested on the real 2023-24 table, the forecast for all 20 clubs missed by 8.4 points on average (correlation 0.88). It had United at 67 points with the real window, which was too high: they got 60.

Window	Expected points	80% range (points)	Average finish	Top 4	Title	Points with real 2023-24 values
Standing still	69	56-83	5.1	48%	7%	60-61
Real window	67	54-81	5.6	40%	5%	60-60
Scenario A: best value	75	62-88	3.9	66%	15%	65-69
Scenario B: most points	82	70-94	2.6	87%	34%	77-78

Table 3. United's 2023-24 under each window. League points, chances from the simulator as seen on 1 September 2023. The last column is the hindsight grade from section 4. 2023-24 had four Champions League places.

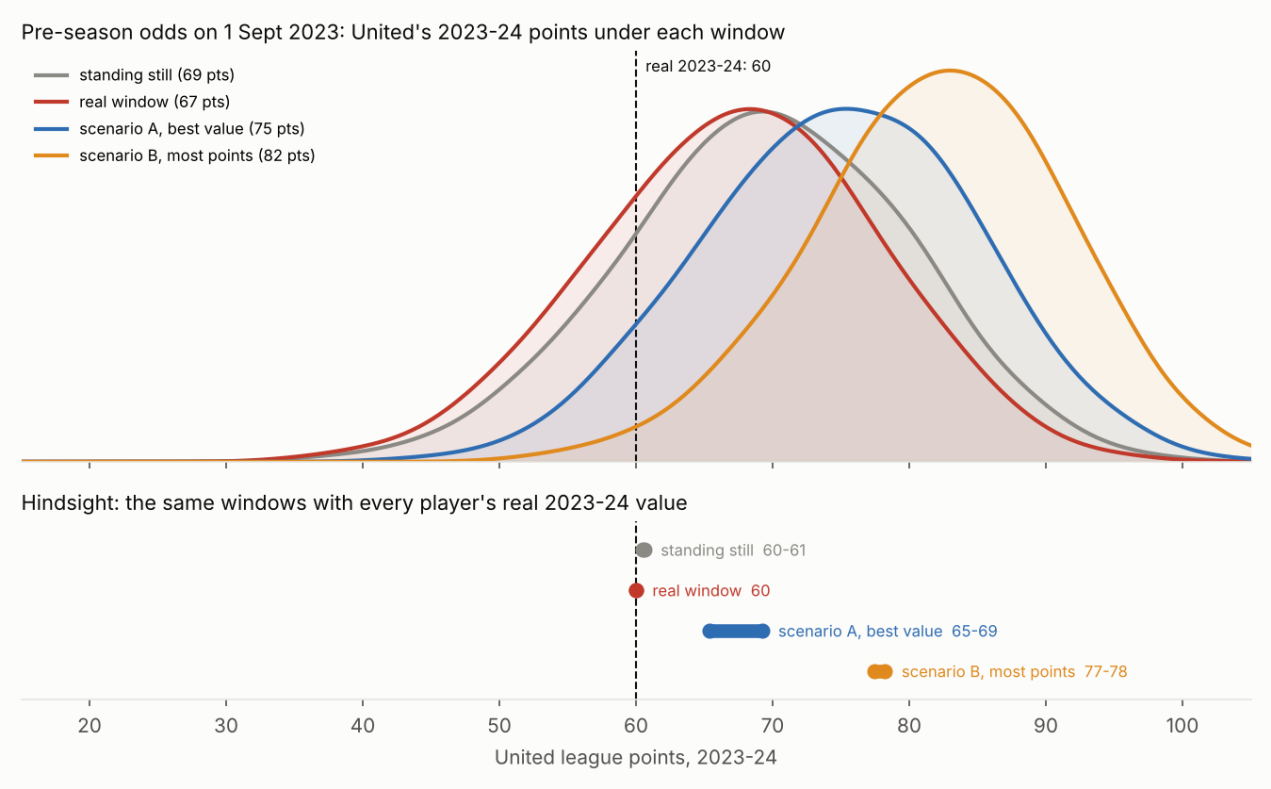


Figure 1. Top: United's simulated 2023-24 league points under each window, as seen on 1 September 2023. Bottom: the same windows scored with every player's real 2023-24 value. The dashed line is the real 60.

The curves in Figure 1 are United's points over the 20,000 simulated seasons. Scenario B's sits about 15 points to the right of the real window's, the difference between a 40% and an 87% chance of the top four. The bars below are the hindsight grade from section 4, a range because it is calculated two ways. The real window sits on 60 because every grade is measured from it.

4. With hindsight

2023-24 is in the data, so every player in every window has a real value for that season: the same RAPM measure, plus shot-stopping for keepers and the league strength adjustment, fitted at the end of 2023-24. Each window is then scored twice with those real values.

- Planned minutes: every player gets the minutes the window planned for him in summer 2023. This is the optimistic end, since it assumes nobody missed more time than expected.
- Capped minutes: every player's minutes are capped at the league minutes he really played in 2023-24, anywhere in the big five, and the minutes he didn't play count as played by a replacement-level player. This is the pessimistic end. It counts every missed minute against the window, also when a player was dropped rather than injured (Sancho played 1,051 league minutes after falling out with Ten Hag), and it assumes United had nobody better than replacement level to fill in.

For a sense of scale: United's real 2023-24 squad, every player on his real value and his real league minutes for United, comes to 19 points above replacement, 12th of the 20 Premier League squads. The line from the main report (about 40 points plus 0.96 per point above replacement) turns that into 58 league points, against the 60 United really got. The capped version of the real window (17) is close to that, as it should be, since it is nearly the same squad on nearly the same minutes.

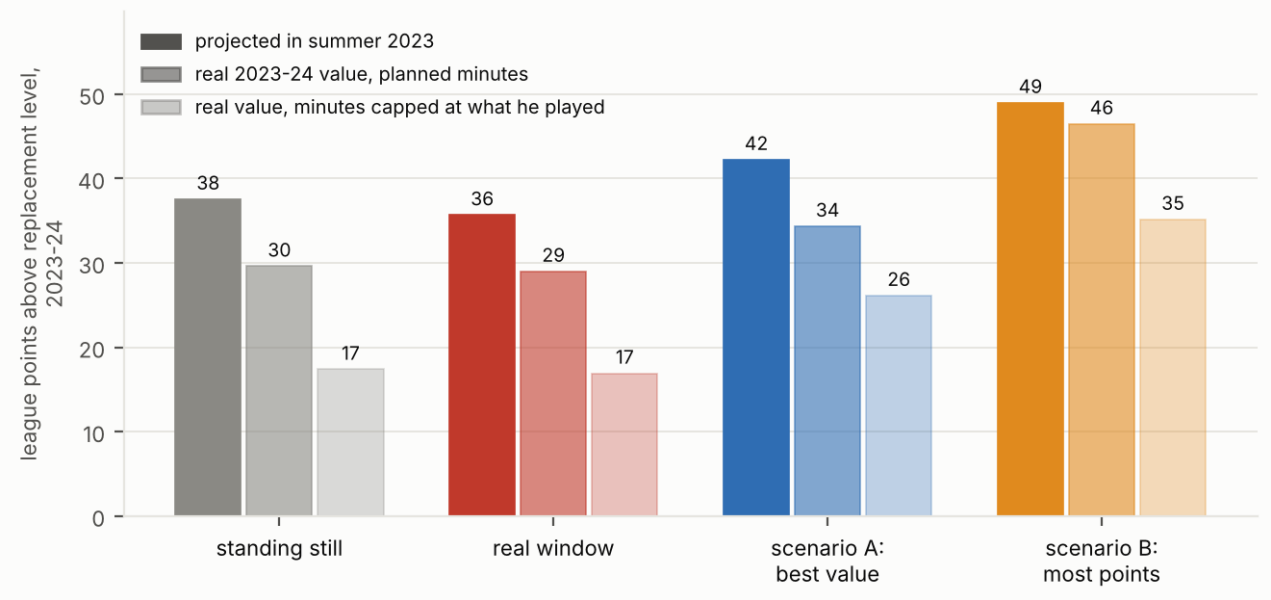


Figure 2. League points above replacement level for 2023-24 per window: as projected in summer 2023 (dark), with the players' real 2023-24 values on the planned minutes (middle), and with their minutes capped at what they really played (light).

Every window drops from the projection to the real values, since the model projected United too high overall: most of the squad rated lower in 2023-24 than on its 2022-23 numbers. The further drop to the capped version is lost minutes. What matters for the grade is the order between the windows, and it holds in both versions. Scenario B keeps 46 of its 49 projected points above replacement, the real window 29 of 36. In the capped version B loses another 11, mostly because Solly March was out with a knee injury from October (571 league minutes) and Botman played 1,383.

The hindsight points in Table 3 come from these bars. Each window's points are United's real 60, moved by its difference to the real window in the same version. For Scenario B: 60 + (46 − 29) = 77 with planned minutes, and 60 + (35 − 17) = 78 with capped minutes (rounded numbers in the sums). Measuring from the real window keeps whatever the values miss, the manager and the luck, the same for every window.

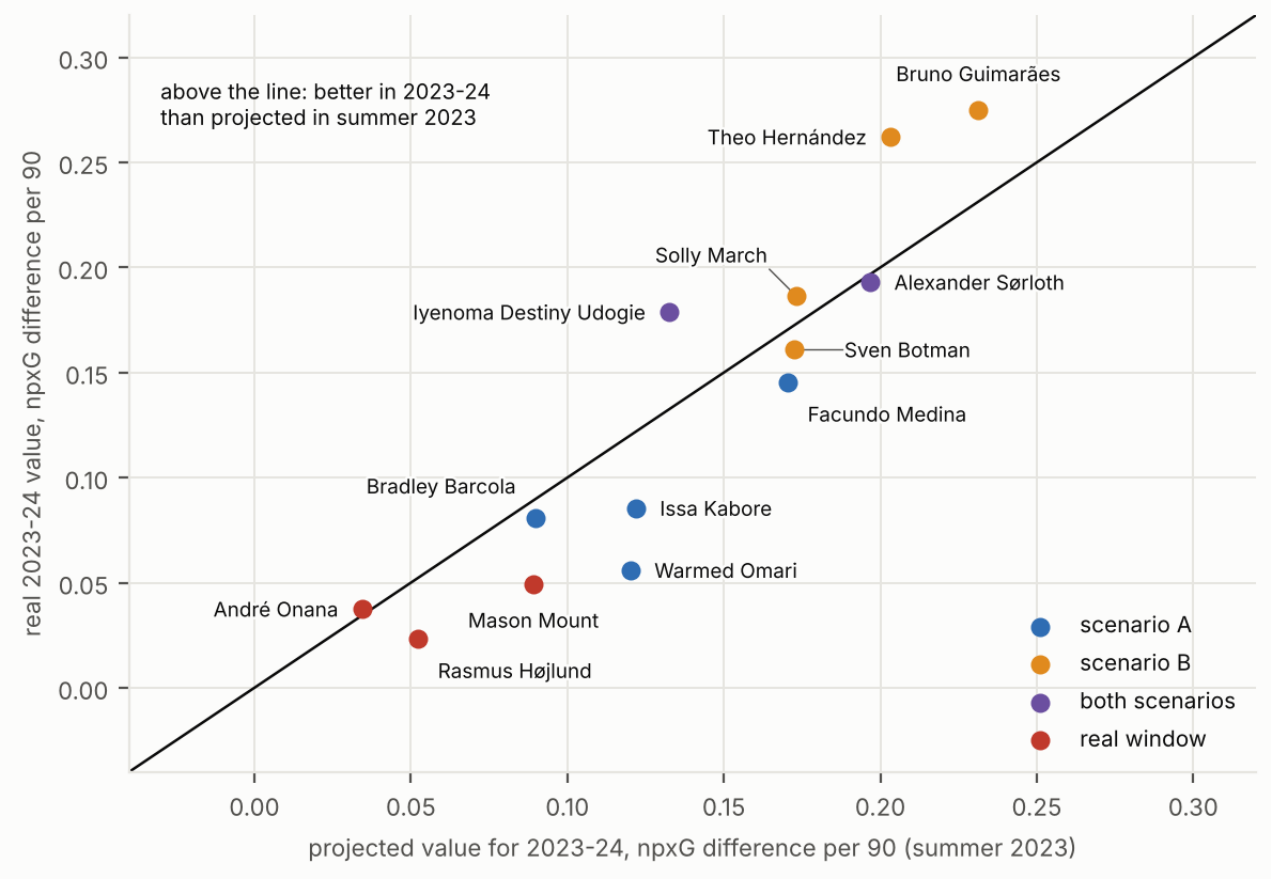


Figure 3. Every signing in the three windows: projected value for 2023-24 made in summer 2023 against the real 2023-24 value, both in npxG difference per 90 against an average player. On the diagonal the projection was exactly right.

Dots above the diagonal were better in 2023-24 than projected. Most of B's signings sit on or above it, for example Bruno Guimarães (+0.23 projected, +0.28 real) and Theo Hernández (+0.20, +0.26). Mount and Højlund fell short of projections that were already low.

Tables 4 to 6 list every move, one row per player, signings (in) first and sales (out) after. Fees are what the window pays: the model's price for the scenarios, the real fee for the real window. Projected value is the model's number from summer 2023, real value the 2023-24 number, both in npxG difference per 90 against an average player, so +0.10 is about 2.8 league points over a full season. "no data" means he played no big-five league minutes in 2023-24. League minutes and market value show what really happened to the player in 2023-24 at his real club, so for the scenarios they are what the player did elsewhere, not at United.

	Player	Fee (EUR m)	Projected value, 23-24	Real value, 23-24	League minutes, 23-24	Market value (EUR m), Jun 23 → Sep 24
in	Iyenoma Destiny Udogie	37	+0.13	+0.18	2,398	22 → 45
in	Bruno Guimarães	65	+0.23	+0.28	3,269	60 → 85
in	Sven Botman	58	+0.17	+0.16	1,378	45 → 45
in	Solly March	17	+0.17	+0.19	559	18 → 18
in	Theo Hernández	64	+0.20	+0.26	2,795	60 → 60
in	Alexander Sørloth	13	+0.20	+0.19	2,491	12 → 25
out	Raphaël Varane		+0.06	+0.10	1,371	40 → 20
out	Anthony Martial		+0.12	+0.05	444	15 → 10
out	Alex Telles		-0.08	no data	0	12 → 6
out	Donny van de Beek		+0.02	-0.00	359	17 → 5
out	Tyrell Malacia		+0.09	no data	0	22 → 18
out	Mason Greenwood		-0.00	+0.03	2,668	50 → 25
	Signings, total paid	253				217 → 278 (+61)
	Sales, total received	125				156 → 84 (-72)
	Net spend	+129				

Table 4. Scenario B, most points: signings (in) and sales (out).

In Table 4, the sales are mostly players who were leaving or barely played: Varane and Martial were in the last year of their contracts, Telles, Van de Beek and Malacia played 359 league minutes between them in 2023-24, and Greenwood was loaned out in reality too. B's money goes on three starters (Guimarães, Theo Hernández, Botman), with Udogie as left-back cover and March and Sørloth as cheaper starters on the right and up front.

	Player	Fee (EUR m)	Projected value, 23-24	Real value, 23-24	League minutes, 23-24	Market value (EUR m), Jun 23 → Sep 24
in	Bradley Barcola	13	+0.09	+0.08	1,647	7 → 50
in	Facundo Medina	22	+0.17	+0.15	2,714	16 → 25
in	Iyenoma Destiny Udogie	37	+0.13	+0.18	2,398	22 → 45
in	Warmed Omari	12	+0.12	+0.06	2,071	8 → 8
in	Issa Kabore	9	+0.12	+0.09	1,732	5 → 10
in	Alexander Sørloth	13	+0.20	+0.19	2,491	12 → 25
out	Raphaël Varane		+0.06	+0.10	1,371	40 → 20
out	Anthony Martial		+0.12	+0.05	444	15 → 10
out	Marcus Rashford		+0.12	+0.00	2,278	80 → 60
out	Aaron Wan-Bissaka		+0.13	+0.06	1,782	22 → 20
out	Lisandro Martínez		+0.07	+0.08	648	50 → 50
out	Antony		+0.09	+0.18	1,323	70 → 25
	Signings, total paid	105				70 → 163 (+93)
	Sales, total received	222				277 → 185 (-92)
	Net spend	-116				

Table 5. Scenario A, best value: signings and sales.

Table 5 shows where A's money comes from: it sells Rashford (EUR 80m market value), Antony (70m), Martínez (50m) and Varane (40m), and buys six players for EUR 105m in total. Antony rated much better in 2023-24 than A expected (+0.18 against +0.09), so that sale cost points. The window still makes money because the wages saved on the four are large.

	Player	Fee (EUR m)	Projected value, 23-24	Real value, 23-24	League minutes, 23-24	Market value (EUR m), Jun 23 → Sep 24
in	Altay Bayındır	5	-0.01	no data	0	13 → 10
in	Rasmus Højlund	80	+0.05	+0.02	2,171	35 → 65
in	André Onana	50	+0.03	+0.04	3,420	20 → 35
in	Mason Mount	68	+0.09	+0.05	514	65 → 35
out	Fred		+0.09	no data	0	20 → 17
out	Alex Telles		-0.08	no data	0	12 → 6
out	Dean Henderson		+0.06	+0.03	1,620	22 → 12
out	Eric Bailly		-0.07	+0.10	838	8 → 3
out	Brandon Williams		-0.01	no data	0	10 → 2
out	Mason Greenwood		-0.00	+0.03	2,668	50 → 25
out	Teden Mengi		-0.01	-0.07	2,470	2 → 12
out	Matej Kovar		-0.01	-0.05	90	1 → 7
out	Anthony Elanga		+0.08	+0.12	2,434	25 → 22
out	Zidane Iqbal		-0.02	no data	0	1 → 1
out	Álvaro Carreras		-0.01	+0.06	473	7 → 7
	Signings, total paid	203				133 → 145 (+12)
	Sales, total received	61				158 → 114 (-43)
	Net spend	+142				

Table 6. The real window. The loans in (Amrabat, Reguilón) and Evans on a free aren't paid moves in Transfermarkt, so they are left out.

Table 6 is the real window. United paid EUR 203m for four players, of whom only Onana and Højlund played a full season, and sold mostly fringe players for EUR 61m.

The market's verdict a year later points the same way. By September 2024 the market value of Scenario A's signings had gone up by EUR 93m (Barcola alone from EUR 7m to 50m), Scenario B's by EUR 61m and the real window's by EUR 12m, with Mount down from EUR 65m to 35m. The players United really sold or loaned out lost EUR 43m in market value; the ones A and B would have sold lost EUR 72m to 92m, so the model sold at about the right time.

5. What it says, and what it can't

The case supports two things. First, the model's values carry real information about the next season for the players it would have bought: Scenario B's signings rated as high or higher in 2023-24 as projected, and the real window's didn't. Second, with the same net spend and rules, Scenario B would have been worth about 17 league points more than the real window. That is roughly the gap between United's real 8th place and a top-four finish in 2023-24 (Aston Villa were 4th on 68).

- Hindsight scores what the players delivered, not the season. Fit, the manager, how a dressing room takes six new players and injuries spreading through the squad aren't in it, and United's real season had a lot of all four.
- The real 2023-24 value isn't purely 2023-24. RAPM pools a season with the three before at lower weights, so a player's earlier seasons still make up about a third of it.
- Scenario B buys four Premier League players (from Newcastle, Spurs and Brighton), Theo Hernández from Milan and Sørloth, who was on loan at Real Sociedad. The seller rule keeps it to clubs rated below United, but whether Newcastle would have sold Guimarães and Botman to a direct rival at the market price is a real question. A sale probability fitted on past transfers would make that uncertainty part of the value; here it is a yes/no rule.
- Scenario A is the textbook moneyball window, and its weakness shows: some of the cheap players it rates highly were noise (Omari +0.12 projected, +0.06 real). Maximising money alone keeps choosing uncertain

bargains, while maximising points with a money floor gives the better window.

- The greedy search makes one move at a time, so it can miss a pair of moves that only works together. With six moves and 133 targets an exact search wasn't needed to make the point.